DSA0601 Data Handling and Visualization

Slot A

CO2 – Assessment Test 1 (AT1)

Chart Construction Test

Instructions:

For each question, write the program in the specified language (R or Python), generate the required visualization, and answer the interpretation sub-questions clearly with justification.

DSA06 – Data Handling and Visualization | Assessment Tool: Experiment

DSA06 – DATA HANDLING AND VISUALIZATION

CO3_AT1_Assessment Tool: EXPERIMENT

Topics Covered: ECDF and Q-Q Plots; Multiple Distributions; Vertical-Axis Distributions; Bean Plots; Proportions; Pie/Side-by Side/Stacked Bars; Stacked Densities; Treemaps; Sunburst Charts; Parallel Sets; Sankey Diagrams; Nested Proportions; Avoiding Misleading Proportions.

General Instruction: For each experiment, use an appropriate visualization, justify the design, interpret the result, and identify potential visualization bias. R/ggplot2 or an equivalent visualization tool may be used.

Question 1 – Distribution Diagnosis using ECDF and Q-Q Plot

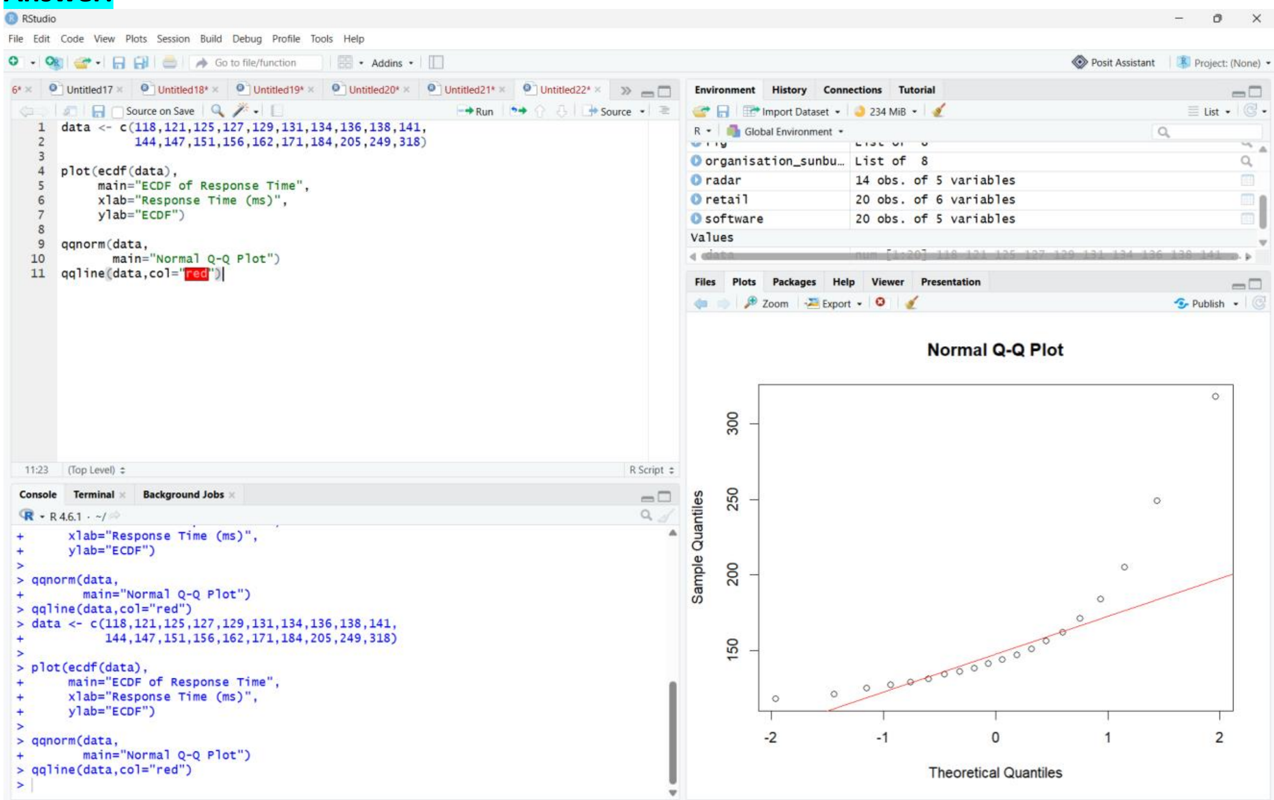
Scenario: An engineering department records response time (ms) of an automated fault-detection system under 20 test runs. The team claims the distribution is approximately normal and wants to use mean and standard deviation for monitoring.

Dataset (n=20): 118, 121, 125, 127, 129, 131, 134, 136, 138, 141, 144, 147, 151, 156, 162,

171, 184, 205, 249, 318 **Tasks:**

1. Construct an ECDF and a normal Q-Q plot.

Answer:



- ECDF displays the cumulative proportion of response times.
- Normal Q-Q plot compares the sample distribution with a theoretical normal distribution.

2. Use both plots to evaluate the normality claim.

Answer:

The response times are not normally distributed.

Justification:

- ECDF shows a long upper tail.
- Q-Q plot deviates from the straight reference line, especially at higher values.
- Large observations (205, 249, 318) pull the distribution away from normality.

3. Identify evidence of skewness and/or extreme observations.

Answer:

The dataset is positively (right) skewed.

Evidence:

- Most values lie between 118–171 ms.
- Few very large values exist:
  - 205 ms
  - 249 ms
  - 318 ms
- These are potential outliers**.**

4. Recommend percentile-based limits or mean ± standard-deviation limits, with justification.

Key competency: ECDF, Q-Q plot, skewness, outliers, distribution interpretation

Answer:

Recommended: Percentile-based limits.

Example:

- 5th Percentile
- 95th Percentile

Justification

Because the data contains:

- right skewness
- extreme observations
- non-normal distribution

Mean ± Standard Deviation assumes normality and may be misleading.

Therefore, percentile limits provide a more robust monitoring method.

Interpretation

- Majority of response times are clustered around **130–170 ms**.
- A few unusually high response times indicate occasional system delays.
- The normality assumption is violated.
- Percentile-based monitoring is preferable to mean ± SD.

Question 2 – Comparing Many Distributions and Bean Plots

Scenario: A semiconductor manufacturer measures chip fabrication cycle time (minutes) for four production lines. Management wants to identify differences in variability and distribution shape.

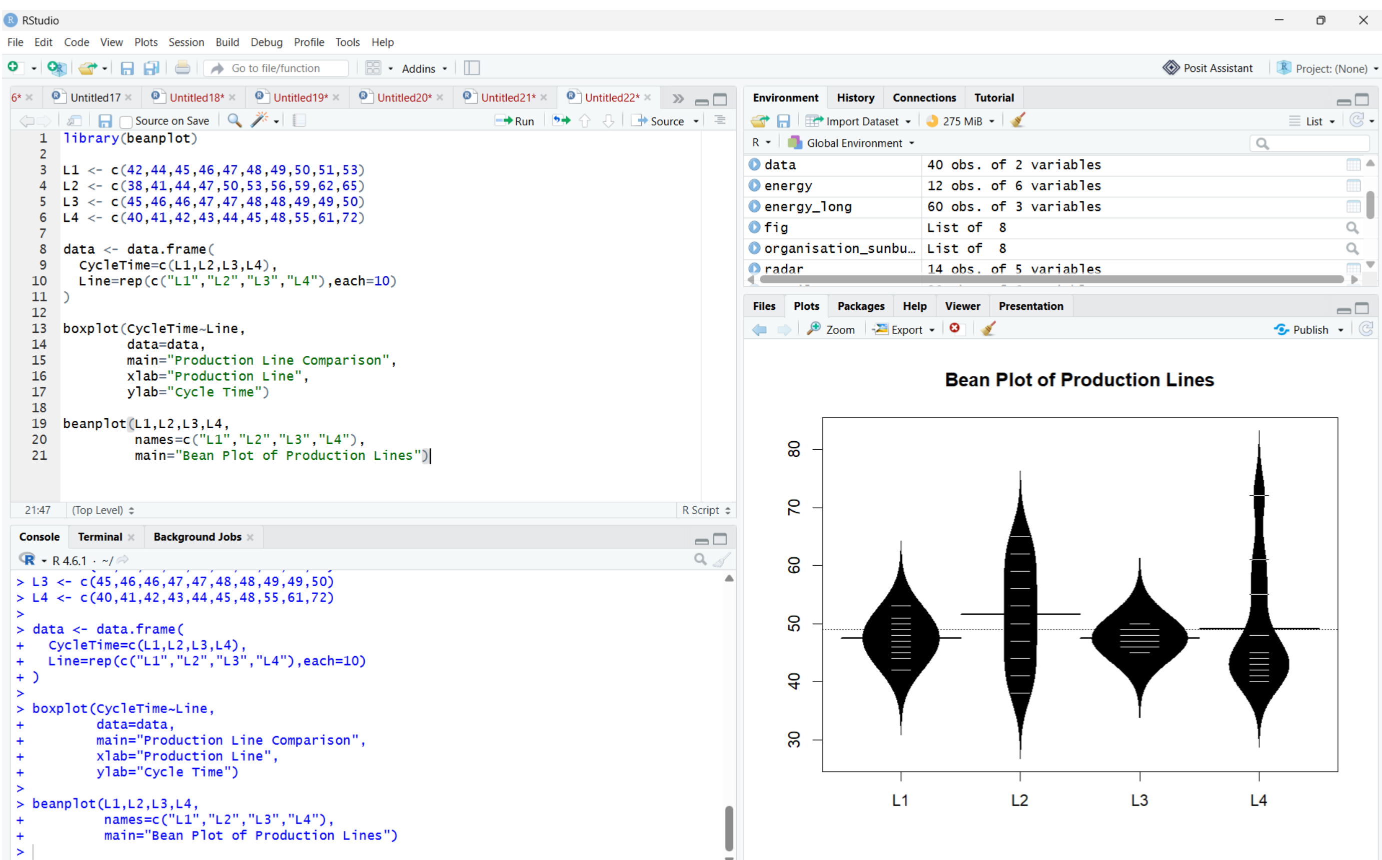
Production Line	**Cycle Times (minutes)**
L1	42, 44, 45, 46, 47, 48, 49, 50, 51, 53
L2	38, 41, 44, 47, 50, 53, 56, 59, 62, 65
L3	45, 46, 46, 47, 47, 48, 48, 49, 49, 50
L4	40, 41, 42, 43, 44, 45, 48, 55, 61, 72

Tasks:

1. Create a visualization comparing all four distributions.

Answer:

A box plot compares the distribution, median, spread, and potential outliers of all four production lines. A bean plot (or violin plot) shows the density and overall shape of each distribution.



2. Construct bean plots or an equivalent distribution visualization along the vertical axis.

   Answer:

   A bean plot displays:

- Data distribution
- Density of observations
- Individual data values
- Median and spread

If a bean plot is unavailable, a violin plot is an appropriate equivalent.

3. Determine which line has greatest variability and which has the most concentrated distribution.

Answer:

- Greatest variability: L4
    - Values range from 40 to 72 minutes, showing the widest spread.
- Most concentrated distribution: L3
    - Values are tightly clustered between 45 and 50 minutes, indicating the least variability.

4. Explain why comparing only means is insufficient.

Key competency: Multiple distributions, vertical-axis distributions, bean plots, variability

Answer:

Comparing only the mean does not reveal:

- Variation in the data
- Distribution shape
- Outliers
- Range of values
- Consistency of production performance

Two production lines may have similar average cycle times but very different variability. Therefore, distribution plots provide a more complete comparison.

Question 3 – Proportion Visualization and Choice of Chart

Scenario: A university analyzes preferred learning mode of 240 students across three departments. The dean wants comparison without exaggerating small differences.

Department	Online (%)	Blended (%)	Face-to-Face (%)
CSE	52	28	20

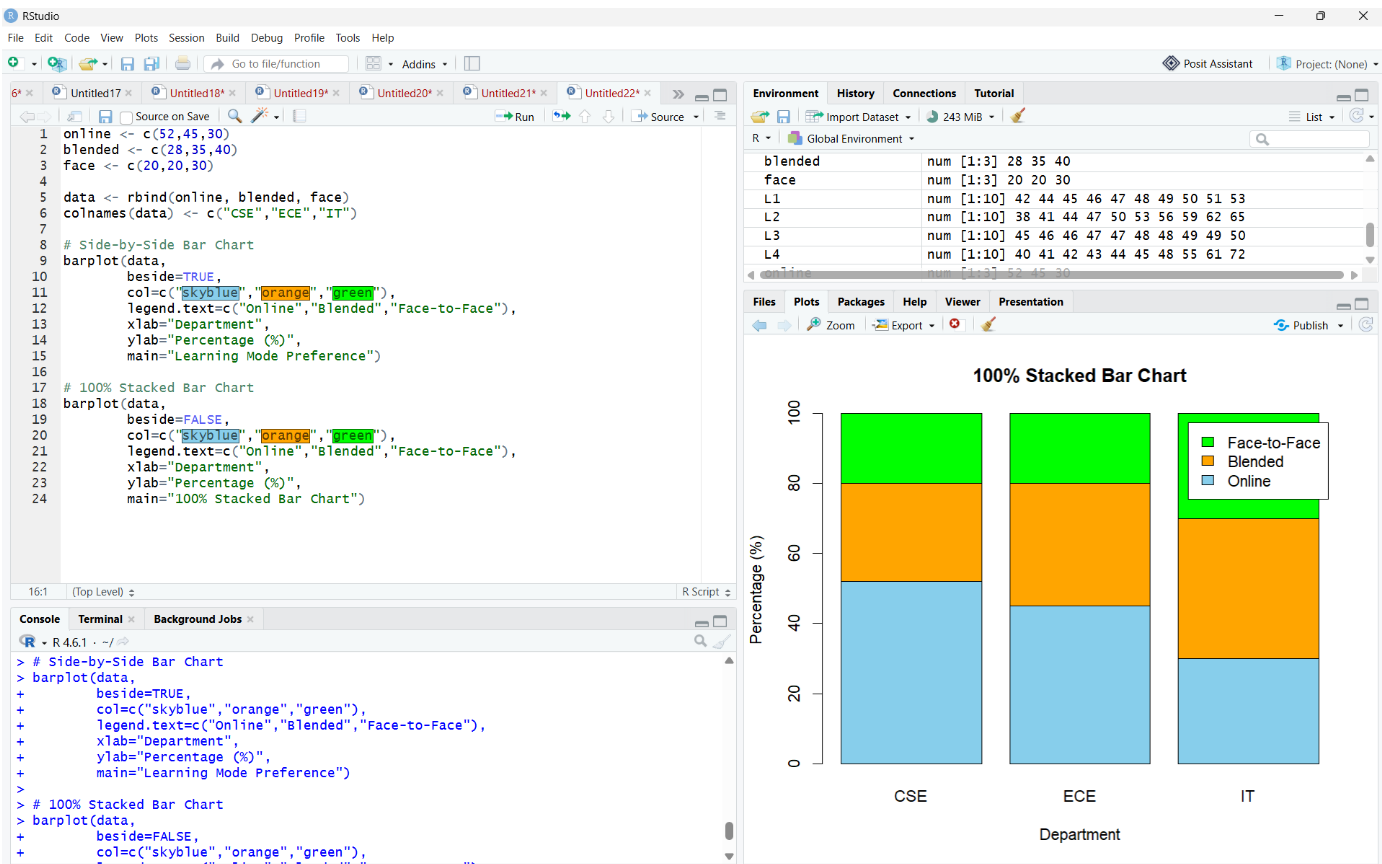
ECE	45	35	20
IT	30	40	30

Tasks:

1. Create a side-by-side bar chart and a 100% stacked bar chart.

Answer:

- Side-by-side bar chart: Displays each learning mode separately for every department, making direct comparison easy.
- 100% stacked bar chart: Shows how each learning mode contributes to the total (100%) within each department.



2. Explain which chart better supports comparison of individual proportions and why.

Answer:

The side-by-side bar chart is better.

Justification:

- All bars start from the same baseline.
- Heights can be compared directly.
- Small differences between departments are easy to identify

3. Explain why separate pie charts may be less effective for comparing departments.

Answer:

Separate pie charts make it difficult to:

- Compare slice sizes across departments.
- Detect small percentage differences.
- Compare multiple categories accurately.

Bar charts provide a common baseline, making comparisons clearer.

4. Identify one design choice that could make small proportional differences appear larger.

Key competency: Proportions, side-by-side bars, stacked bars, pie charts, misleading proportions

Answer:

One misleading design choice is:

- Truncating the y-axis (not starting from zero) in a bar chart.

This exaggerates small differences and can give a false impression of variation.

Question 4 – Nested Proportions using Treemap and Sunburst

Scenario: An e-commerce company analyzes annual revenue of ₹1.2 crore across product categories and subcategories. The dashboard must show both category contribution and internal composition.

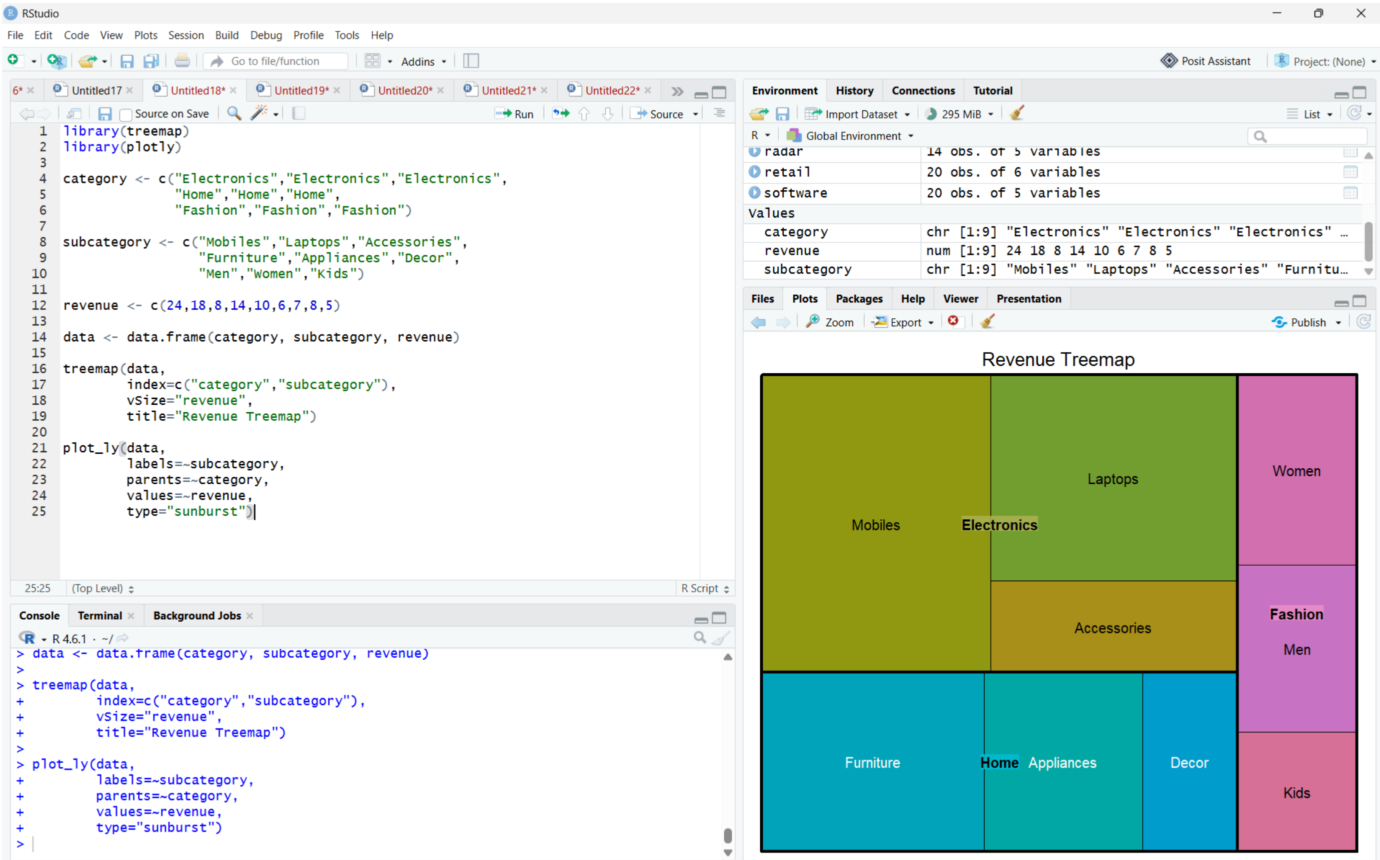
Category	**Subcategory**	**Revenue (₹ lakh)**
Electronics	Mobiles	24
Electronics	Laptops	18
Electronics	Accessories	8
Home	Furniture	14
Home	Appliances	10
Home	Decor	6
Fashion	Men	7
Fashion	Women	8
Fashion	Kids	5

Tasks:

1. Construct a treemap showing category and subcategory revenue.

Answer:

A Treemap represents categories and subcategories using nested rectangles. The area of each rectangle is proportional to revenue. Larger rectangles indicate higher revenue contributions.



2. Construct a sunburst chart representing the same hierarchy.

Answer:

A **Sunburst Chart** displays hierarchical relationships using concentric circles:

- Inner circle → Categories
- Outer circle → Subcategories

The size of each segment corresponds to revenue.

3. Determine the largest category and calculate its percentage of total revenue.

Answer:

Category Revenue:

Electronics

= 24 + 18 + 8 = **50 lakh**

Home

= 14 + 10 + 6 = **30 lakh**

Fashion

= 7 + 8 + 5 = **20 lakh**

Total Revenue = **100 lakh**

Percentage of Electronics: Percentage = 100/50 ×100

Largest Category: Electronics (50%).

4. Compare treemap and sunburst for identifying nested proportions.

Answer:

Treemap	**Sunburst**
Uses rectangular areas	Uses circular segments
Efficient use of space	Better hierarchy visualization
Easier area comparison	Easier parent-child relationship understanding
Good for exact comparison	Good for structure analysis

Conclusion:

- Treemap is better for comparing values.
- Sunburst is better for showing hierarchy.

5. State one situation in which area encoding could mislead interpretation.

Key competency: Treemaps, sunburst charts, nested proportions, area encoding

Answer:

Area encoding can mislead when:

- Very small categories become difficult to compare.
- Human eyes may incorrectly judge area differences.
- Similar-sized rectangles or sectors may appear unequal.

Example:
Two subcategories with revenues of ₹7 lakh and ₹8 lakh may look almost identical, making exact comparison difficult.

Question 5 – Flow Proportions using Sankey Diagram / Parallel Sets Scenario: A smart-city mobility study follows 200 commuters from residential zone through primary transport mode to destination zone. The planning team wants dominant flows and possible bottlenecks.

Origin	**Mode**	**Destination**	**Commuters**
Residential	Bus	Central	35
Residential	Bus	Industrial	15

Residential	Train	Central	25
Residential	Train	Industrial	10
Suburban	Bus	Central	20
Suburban	Bus	Industrial	10
Suburban	Train	Central	30
Suburban	Train	Industrial	15
Rural	Bus	Central	12
Rural	Bus	Industrial	8
Rural	Train	Central	10
Rural	Train	Industrial	10

Tasks:

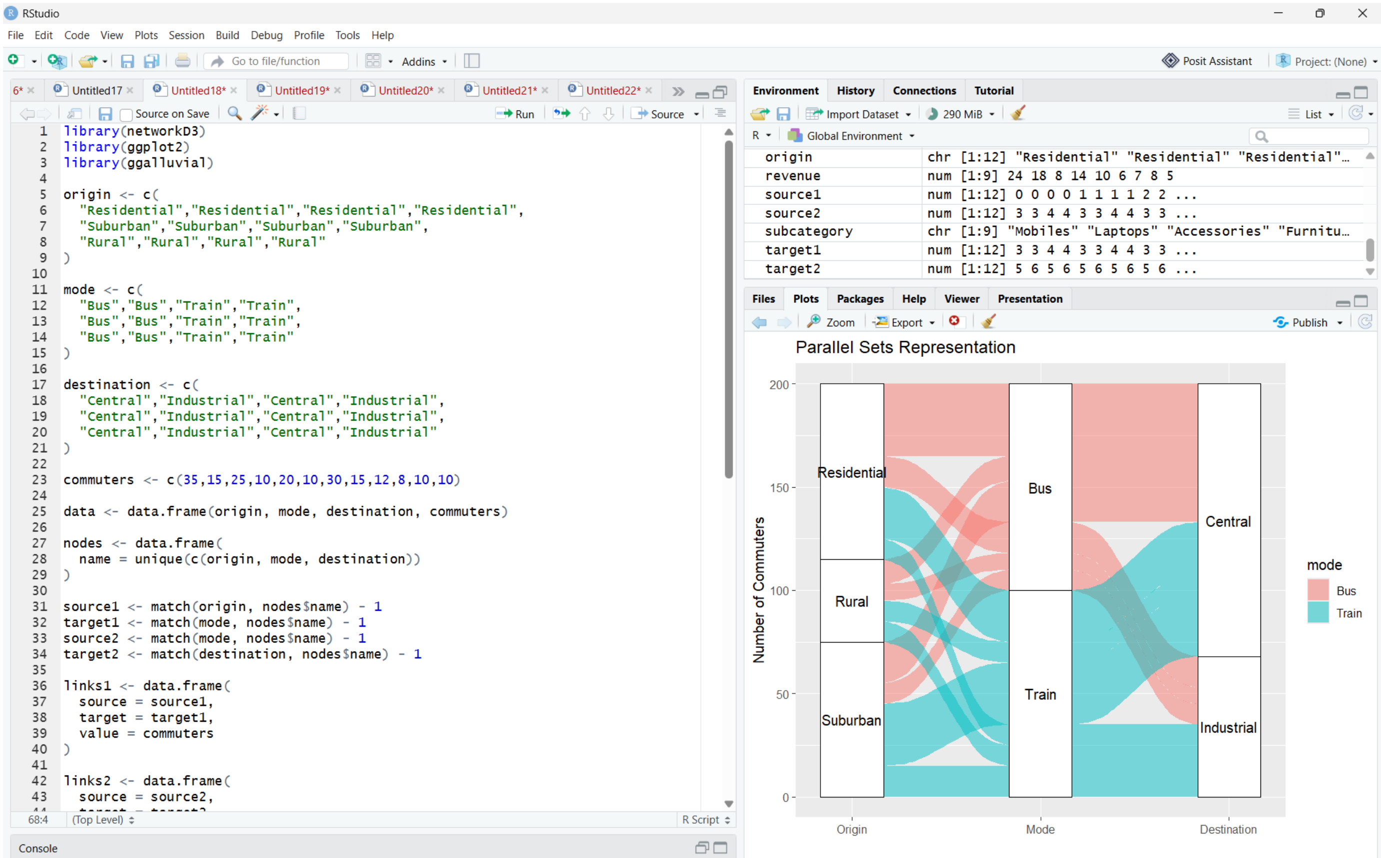
1. Construct a Sankey diagram showing Origin → Mode → Destination.

Answer:

The Sankey diagram represents the flow:

Origin → Transport Mode → Destination

The width of each link represents the number of commuters. A wider link indicates a larger commuter flow.



2. Construct a parallel-sets representation of the same categorical flows.

Answer:

Parallel sets represent the same categorical relationships using connected bands between:

Origin → Mode → Destination

The thickness of each band represents the number of commuters.

3. Identify the three largest flows and calculate their percentages of all commuters.

Answer:

Total commuters:

The three largest flows are:

Rank	Flow	Commuters	Percentage
1	Residential → Bus → Central	35	17.5%
2	Suburban → Train → Central	30	15%
3	Residential → Train → Central	25	12.5%

Percentage formula:

Percentage= 200/Flow ×100

Therefore, the three largest flows represent **17.5%, 15%, and 12.5%** of all commuters.

4. Compare Sankey and parallel sets for flow magnitude and category relationships.

Answer:

Sankey Diagram	Parallel Sets
Best for showing flow magnitude	Best for categorical relationships
Link width clearly represents quantity	Band width represents quantity
Easy to identify dominant flows	Easy to compare multiple categories
Suitable for flow/bottleneck analysis	Suitable for multidimensional categorical data

5. Explain one way poor link widths or category ordering could mislead the viewer.

Key competency: Sankey diagrams, parallel sets, flow proportions, categorical relationships

Answer:

Poorly chosen **link widths** can make flows appear larger or smaller than their actual values.

Also, poor **category ordering** can create unnecessary crossings and make relationships difficult to follow.

Suggested Evaluation Rubric – Experiment

Criterion	Excellent (4)	Good (3)	Satisfactory (2)	Needs Improvement (1)
Data preparation	Correct and complete	Minor issue	Several issues	Incorrect/incomplete
Visualization choice	Highly appropriate and justified	Appropriate	Partially appropriate	Poor choice
Technical execution	Accurate and clear	Mostly accurate	Some errors	Major errors
Interpretation	Insightful and evidence-based	Correct	Basic	Incorrect/unsupported
Misleading-design analysis	Bias clearly identified and explained	Bias identified	Limited explanation	Not identified

Suggested marks: 20 marks per question; 100 marks total. Level: Competitive / advanced undergraduate.